Statistical Analysis with Missing Data Using Multiple Imputation
Lecture 6
MI with prognostic models

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2024

Prediction Models

Sample size

Development of a prediction model

Validation

Conclusions

Extra material

Timeline

- 2:00-3:15 Lecture
- 3:15-3:30 Break
- 3:30-5:00 Practical

Acknowledgements

With thanks to Prof Ruth Keogh (LSHTM) and Dr Tim Morris (UCL)

Prediction Models

What is a prediction model?

A prediction model aims to use various characteristics about a patient in order to predict whether they have (or will have) an outcome of interest within a given time.

Examples could include

- Diagnosing whether a patient will develop a type of cancer
- Evaluating the risk of a patient dying within 90 days of receiving a surgery

In the medical literature, many prediction models are developed using regression-based models (linear, logistic, proportional hazards regression).

Prediction model example: QRISK + MI

QRISK [1] is a cardiovascular disease risk score developed on 1.28 million patients in the UK and validated on 0.61 million.

Covariates included the likes of sex, BMI, smoking status, total cholesterol and family history of coronary heart disease.
Some covariates, including cholesterol had missing values and the authors used MI.

Initially, the authors did not include the cardiovascular disease outcome in the imputation model which weakened the association between cardiovascular disease and cholesterol [2].

Once the outcome was included, alongside other relevant covariates such as prescriptions of statins, the association was increased to more plausible values and closer to the complete-case result.

Steps in prediction modelling

When developing and evaluating a prediction model, some of the key things to consider are:

- Sample size for model development
- Development of prediction model
  - Variable selection
  - Functional form
  - Parameter tuning (Ex. the choice of α in Lasso regression)
- Validation of prediction model
  - Internal validation
  - External validation
- Deployment of a prediction model to new patients

Sample size

Sample size calculations

To build a *reliable* prediction model, the size of the dataset used to train it needs to be sufficiently large.

For binary or time-to-event outcomes, there used to be a rule of thumb that 10 events were needed for each regression parameter that was being considered for inclusion into the prediction model. But the required no. of events may actually be context specific, relying not just on the no. of parameters but also the number of patients, the outcome proportion and the expected performance of the model [3].

Sample size calculations

Updated guidance on sample size for development of prediction models includes

- Continuous outcomes (Riley et al. 2019 [4])
- Binary outcome (van Smeden et al. 2019 [5] or Riley et al. 2019 [6])
- Time-to-event outcomes (Riley et al. 2019 [6])
- External validation (Snell et al. 2021 [7])

The `pmsampsize` [8] package in R implements the Riley sample size rules.

Sample size Criteria

For binary outcomes:

- Criteria 1: small overfitting defined by an expected shrinkage of predictor effects by 10% or less
- Criteria 2: small absolute difference of 0.05 in the model's apparent and adjusted Nagelkerke's R-squared value
- Criteria 3: precise estimation (within +/- 0.05) of the average outcome risk in the population for a key timepoint of interest for prediction

Sample size example

```
pmsampsize(type="b", cstatistic  = 0.17, parameters = 2,
                   shrinkage = 0.9, prevalence = 0.3)
```

```
Given input C-statistic = 0.17  & prevalence = 0.3
Cox-Snell R-sq = 0.2744

NB: Assuming 0.05 acceptable difference in apparent & adjusted
    R-squared
NB: Assuming 0.05 margin of error in estimation of intercept
NB: Events per Predictor Parameter (EPP) assumes prevalence=0.3

Samp_size Shrinkage Parameter CS_Rsq Max_Rsq Nag_Rsq   EPP
Criteria 1        55     0.900         2 0.2744   0.705   0.389
Criteria 2        48     0.886         2 0.2744   0.705   0.389
Criteria 3       323     0.900         2 0.2744   0.705   0.389
Final            323     0.900         2 0.2744   0.705   0.389

Minimum sample size required for new model development based on
user inputs = 323, with 97 events (assuming an outcome
prevalence = 0.3) and an EPP = 48.45
```

Sample size calculations

The sample size calculations referenced above assume that your data will be fully-observed.

As we know from trial sample size calculations, we may need to make adjustments to allow for the fact that we will have missing values present in our data.

For example, in clinical trials the estimated sample size n can be adjusted for a complete-case analysis (assuming MCAR) by inflating the sample size by the proportion of patients p who are expected to be partially-observed ($n * \frac{1}{1-p}$). It is not clear that this approach would be appropriate for risk prediction, since the information loss for measures of predictive performance is not the same.

Further work is needed for sample size calculation accounting for missing data in a prediction setting.

Development of a prediction model

Missing data and prediction models

Missing data complicate the development of clinical prediction models, determining their performance, and their use to obtain predictions for new patients. The stages are:

1. Handling missing data when training a clinical prediction model
2. Handling missing data while validating the prediction model
3. Handling missing data in new patients when applying the prediction model

Stage 1 - well-researched, including variable selection, functional forms of continuous covariates and other considerations of model development.

Stage 2 and 3 - research ongoing. There is some uncertainty on the best way to handle missing data here.

Ideal and Pragmatic scenarios

Wood et al. [9] investigated the use of MI when using a prediction model to make predictions and evaluate the model's performance. Two estimand-like measures were defined in terms of the practical context in which the predictions would be used for future individuals:

- **Ideal model performance** focuses on an ideal clinical setting where all individuals have fully-observed predictors.
  Essentially, missing data can be thought of as a feature that happens to occur in the data the prediction model will be developed in, but would not occur in future use
- **Pragmatic model performance** is based in a real-world clinical setting where some individuals have missing predictor values.
  It is expected that any future patient data will also have unobserved values in predictors. The handling of missing data in new individuals for whom predictions are to be obtained must be considered in addition to missing data in the dataset used for model development and evaluation.

Stage 1: Training a prediction model

In this lecture we will focus on logistic regression which aims to predict an outcome Y using covariates $\mathbf{X}$ where X_1 contains missing values.

Y			
X_1	NA	NA	NA
X_2			

$$\text{logit}(P(Y_i = 1 \mid \mathbf{X}_i)) = \beta_0 + \sum_{p=1}^{P} \beta_p X_{i,p}$$

Consideration must also be given to the variables to include in the prediction model and the functional form of continuous variables.

Imputation models to fit the prediction model

Using the entire dataset, we are interested in fitting

$$\text{logit}(P(Y_i = 1 \mid \mathbf{X}_i)) = \beta_0 + \sum_{p=1}^{P} \beta_p X_{i,p}$$

but we have missing values in X_1 and want to use multiple imputation to impute these unobserved values.

Per advice in Session 3, we would typically impute X_1 using:

$$X_1 = f(Y, X_{-1}; \boldsymbol{\psi})$$

We include the outcome Y in the model here to maintain the association between the imputed values and the outcome.

Bias in the prediction model regression parameters

While one could argue that in a prediction setting we are less interested in the bias of model parameters, these parameters are used to estimate predicted values for new observations.

Therefore, bias in the parameters will influence the predicted values and we cannot know in practice how large this bias will be.

Previous work focusing on an inference setting for the handling of missing data in covariate selection [10, 11] or non-linear effects [11, 12], as presented in Session 3, can be transferred to the prediction setting.

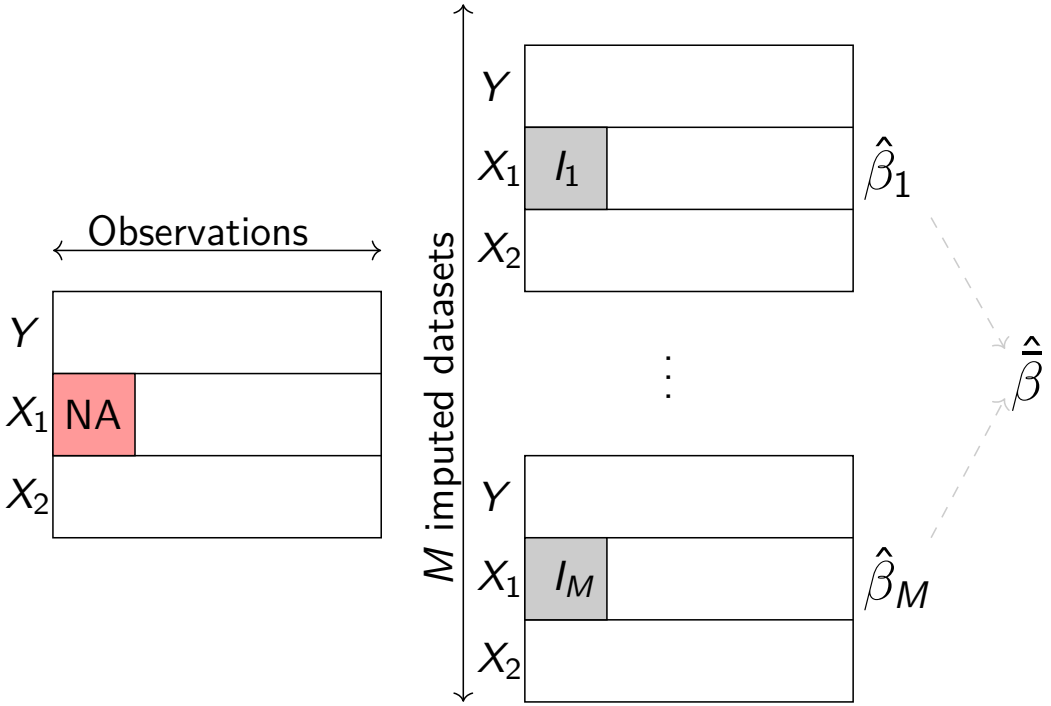
Variable selection with missing data

Using complete-case analysis for variable selection can reduce computational complexity as variable selection is run on only one dataset. However, it can fail to detect important predictors due to loss of power and (when data are not MCAR) bias in Type I error rates.

Simple imputation techniques which involve replacing the missing value with a single imputed value (mean, mode, single regression-based) can also reduce computational complexity as variable selection is run on only one dataset.

However, the choice of imputed value could lead to an unstable model selection process where one choice of imputed value could lead to different covariates selected, than if missing values had been imputed with another value/method.

Variable selection and MI



In MI, the variable selection procedure can be performed separately in the M imputed datasets. Different models can be selected across the M imputed datasets.

Variable selection and MI

Unlike in the inference setting, applying Rubin's rules to M prediction models to get a pooled prediction model may not be required. When training a prediction model and applying it to a new person the options are:

1. Keep the prediction models unpooled and average the predicted values for the individual
2. Pool the prediction model and have one predicted value for the individual

Option 1 can reduce analysis complexity as the analyst does not need to make a decision on combining prediction models with different selections of covariates (or functional forms). Differences between the two options have been noted to be minor [9]

Variable selection and MI

Obtaining one overall analysis model after MI was explored by Wood et al. [10] who considered repeated use of Rubin's rules and 'stacking' imputed datasets for backwards selection.

- *stacking*: The M imputed datasets are combined into one large dataset and backwards selection is applied to the large dataset.
- *R Rules*: The model is fitted in each imputed dataset and then Rubin's rules are applied to get a pooled model. The covariate with the largest (pooled model) p-value is then removed. The reduced model is then fitted in each imputed dataset, and this process is repeated.

Variable selection, functional form and MI

- *stacking*: The parameter estimators are asymptotically equivalent to the averaged parameters from MI Rubin's rules but the standard errors obtained from the stacked data are too small - Wood discusses some weighting approaches to correct this.
- *R Rules*: For backwards selection this was recommended by Wood et al. Note, this involves repeated model-fitting on M imputed datasets and can be computationally intensive.

Additional work by Morris et al. [11] uses multivariable fractional polynomial selection to flexibly model non-linear effects of continuous covariates and variable selection. They found that the stacked approach and Rubin's rules approaches performed similarly in terms of exponent selection.

Functional form

There are various ways to model non-linear effects.

- *Fractional polynomials*: FPs are of the form X_1^E (FP1), $X_1^{E_1} + X_1^{E_2}$, or $X_1^{E_1} + X_1^{E_1} \log(X_1)$ for $E \in \left(-2, -1, -\frac{1}{2}, \log, \frac{1}{2}, 1, 2, 3\right)$. For MI and fractional polynomials, refer to Morris et al. [11]
- *Splines*: There are various types of splines which in general split or window the distribution of the continuous variable. For MI and splines, SMCFCS [13] and R code from Bartlett

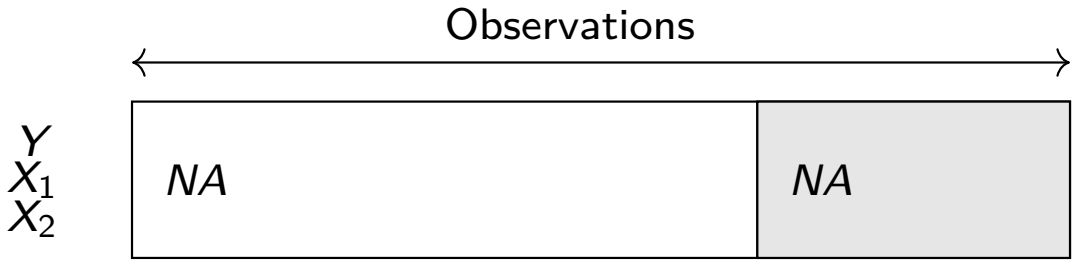
Refer to Lecture 3 notes. For a thorough discussion of handling non-linear effects, refer to [14, Chapter 6]

Validation

Internal validation approaches

Various approaches for internal validation include:

- Apparent performance (evaluate model in the data it was trained in)
- Split into a train and test set
- K-fold Cross-validation
- Optimism-corrected bootstrap



An example of splitting data into a training set (white) and a test set (grey).

Performance Measures for binary outcomes [15]

'Overall Measures'

Measures the distance between the predicted and observed outcome. The Brier score (binary outcome) ranges between 0 and 1. A lower value indicates a better performing model. Brier $= \frac{1}{N}\sum_{i=1}^{N}(\hat{p}_i - y_i)^2$

Discriminative ability

How well a model can differentiate between those who do or do not have the outcome. Area under the receiver operating characteristic curve (AUC) ranges between 0 and 1. A value of 0.5 indicates a model which does not perform any better than chance. A higher value indicates the model is doing better than a random guess.

Performance Measures for binary outcomes [15]

Calibration
A well calibrated model should provide reliable predicted risks which correspond to the observed proportions of the event [16]

i.e. if the predicted risk of an outcome is $r\%$, then we expect $r\%$ of observations to have the outcome at a group-level.

There are several levels of calibration: mean, weak, moderate and strong.

The moderate and strong levels assess calibration using plots which makes assessment in multiply imputed datasets difficult.

Imputation models for training and testing

When implementing in either the ideal or pragmatic setting, the training imputation model should include the outcome and any relevant covariates which will improve the quality of the imputed values.

Inclusion of the outcome will maintain the association between it and the values of the variable being imputed in the training set.

However, multiple imputation of the test set will vary depending on whether ideal or pragmatic performance is of interest.

Test imputation model - ideal scenario

The outcome Y can be included in the test imputation model for imputing the test set in order to maintain the association between outcome and imputed values.

This replicates the scenario of a new patient being fully-observed i.e. there will be no missing values in X_1 and therefore the underlying true data-generating process relating X_1 to Y will be maintained (under the assumption of missing-at-random).

Estimand	Test imputation model	Comments
Ideal	$X_1 = f(Y, X_2; \psi)$	Using observed values of Y when fitting imputation model.

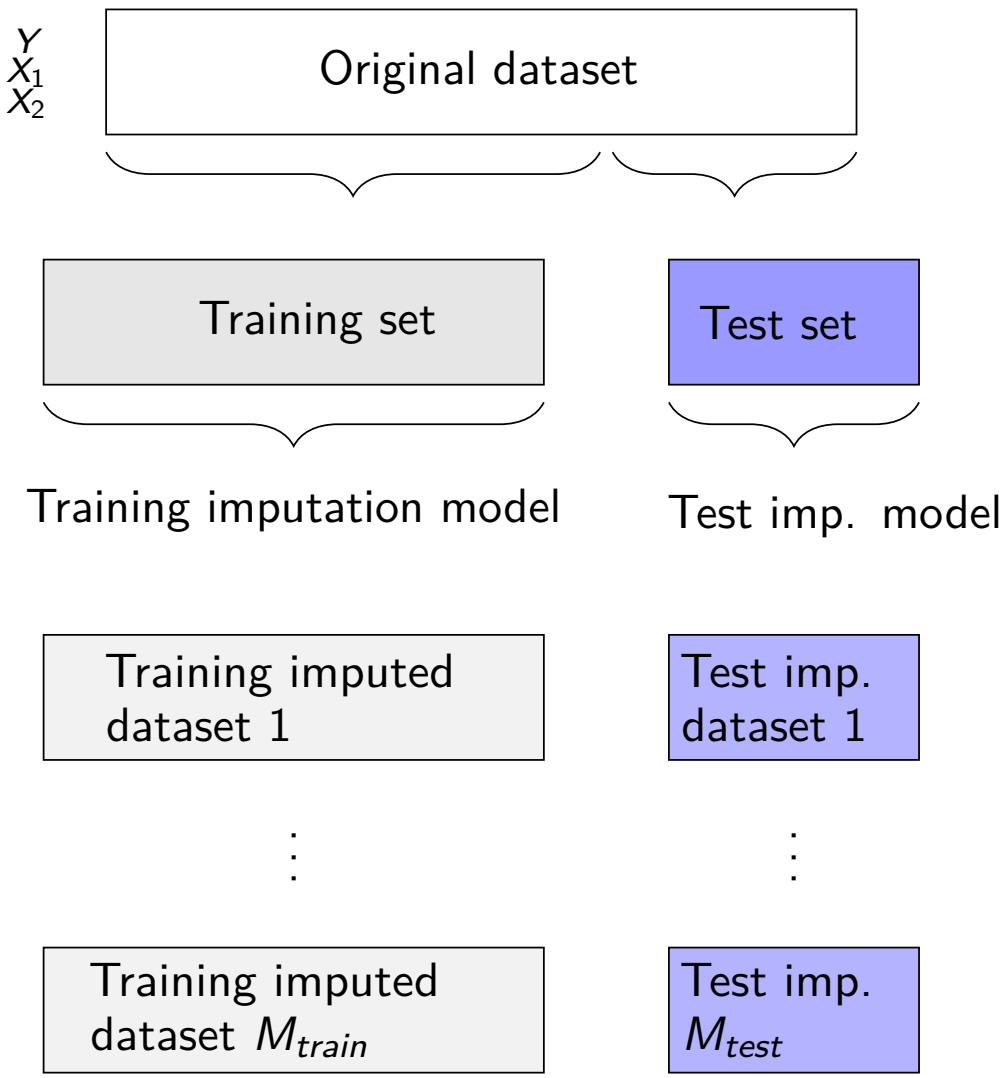
Test imputation model - pragmatic scenario

Future patients are expected to have missing values in X_1 under the same mechanism as in the current data used to train the model. However, in practice these new patients will not have an observed outcome yet.

Below we outline potential imputation models if MI will be used to impute missing values in future patients.

Estimand		Test imputation model	Comments
Pragmatic	A	$Y = f(X_1, X_2; \boldsymbol{\psi}_Y)$ $X_1 = f(Y, X_2; \boldsymbol{\psi})$	Values of Y in test set are set as missing. Both Y and X_1 are imputed. Discard the imputed values of Y.
	B	$X_1 = f(X_2; \boldsymbol{\psi})$	Y removed completely from test imputation model.

An example of training and test imputed datasets



Cross-validation (CV)

Original data

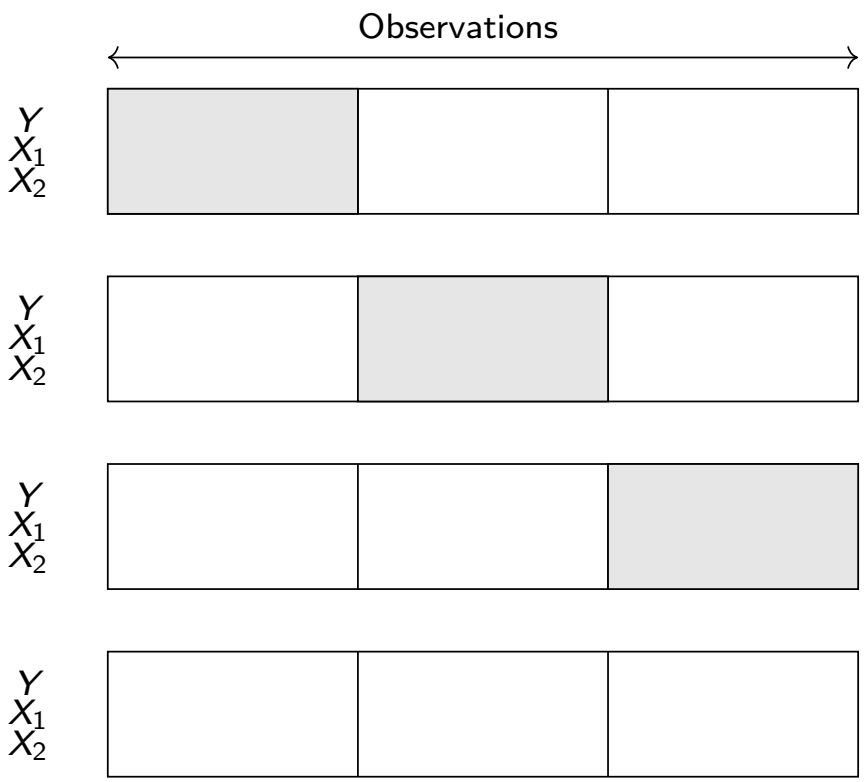
Y X_1 X_2	NA $\qquad\qquad\qquad\qquad$ NA

$$\text{logit}(P(Y_i = 1 \mid \mathbf{X}_i)) = \beta_0 + \sum_{p=1}^{P} \beta_p X_{i,p}$$

- We want to assess the predictive ability of the logistic regression model developed using all of the data
- How will the model perform on 'new' data? To mimic this process, we can split the data into training and test folds.

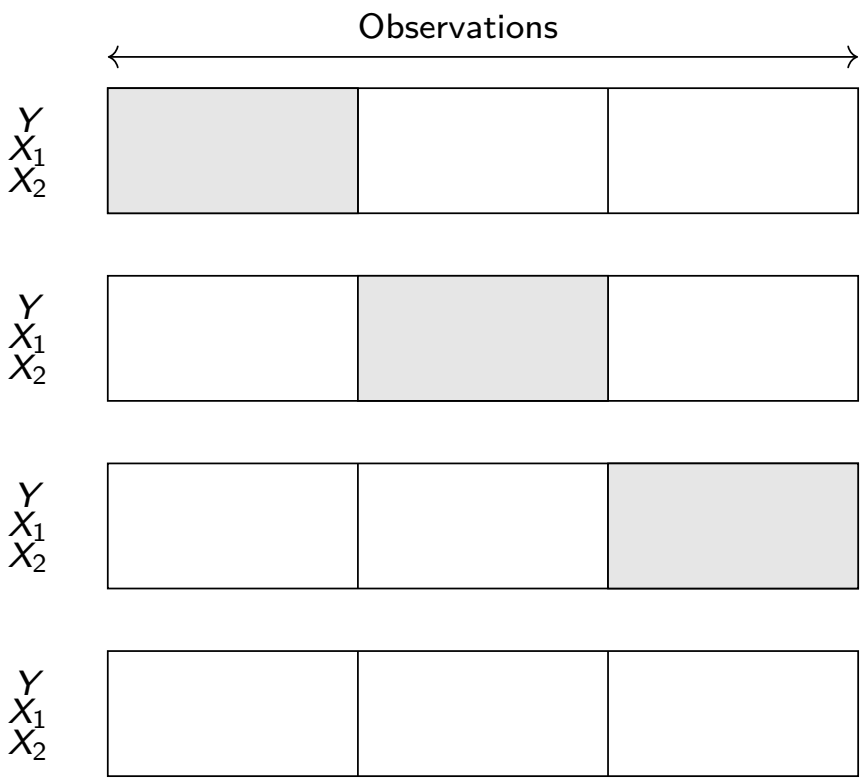
Cross-validation with no missing data

CV is a resampling technique which can be used to evaluate prognostic models. It involves splitting the data into K folds, and training a model on $K-1$ folds (white) and evaluating the model developed on the kth test fold (grey). In the example below, $K = 3$:



Model	Performance
1	Perf_1
2	Perf_2
3	Perf_3
Final	$\text{Perf}_{Overall}$

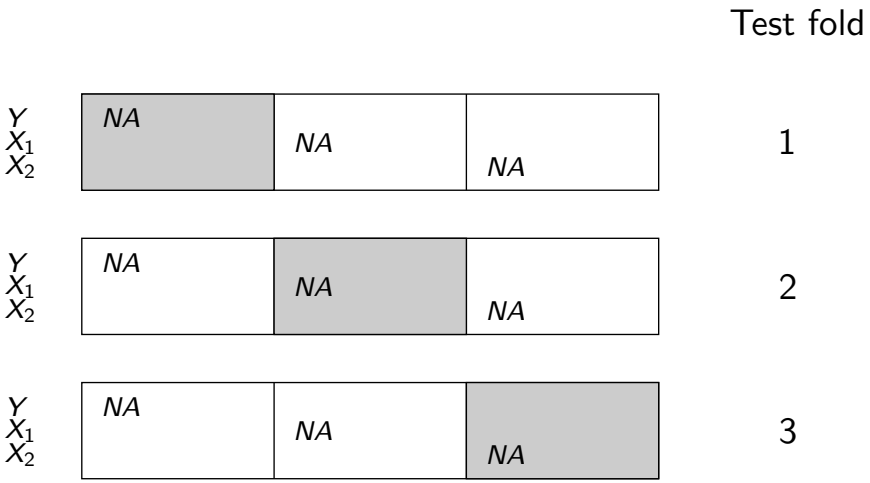
Cross-validation with no missing data



Model	Performance
1	Perf_1
2	Perf_2
3	Perf_3
Final	$\text{Perf}_{Overall}$

- We will have 3 estimates of performance (Perf_k) which we average to get an overall estimate of performance ($\text{Perf}_{Overall}$)
- $\text{Perf}_{Overall}$ is the estimated performance for the model developed using all K folds $\text{Perf}_{Overall} = \sum_{k=1}^{K} \frac{\text{Perf}_k}{K}$

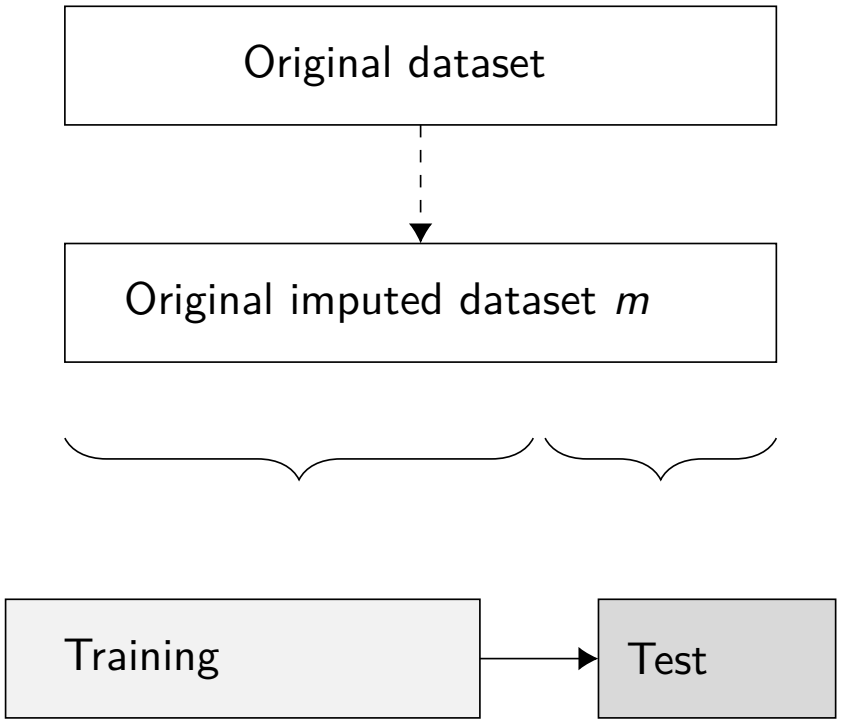
How do we impute when using CV?



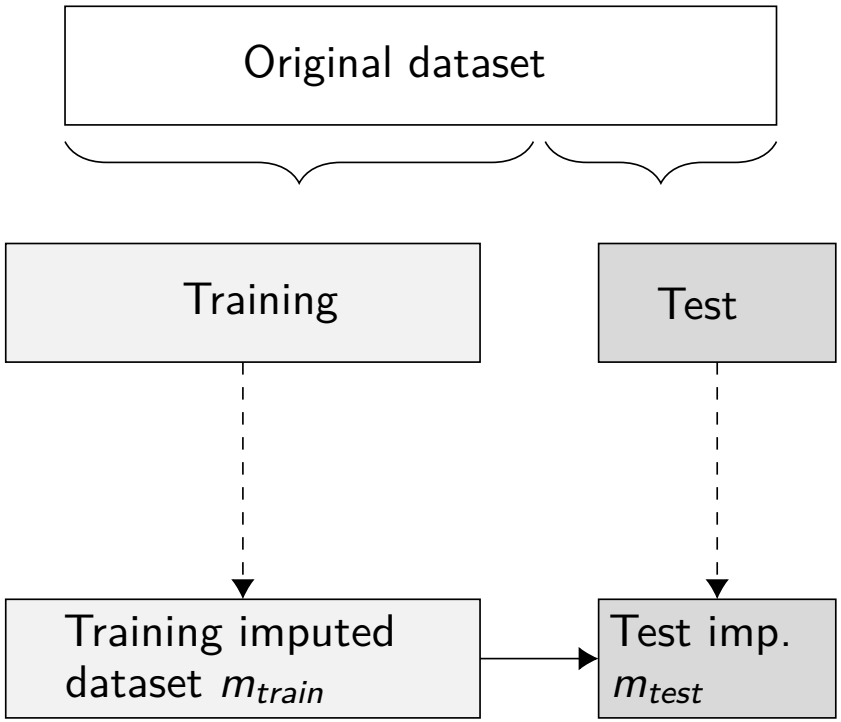
- Impute dataset at the start and then use CV?
- Impute each fold separately?
- Allow holdout data to be included in imputations of training data?
- Allow training data to be included in imputations of holdout data?
- Include the outcome Y?

MI-then-CV (a) or CV-then-MI (b)?

For one iteration of cross-validation:



(a) Imputing missing data followed by splitting into training and test sets



(b) Splitting data into a training and test set and then imputing each separately

Debate in the literature

There is a wide array of literature in this field which often provides conflicting advice. Much of the literature has focused on the ideal performance estimand.

MI-then-Validate

- Wood et al. 2015 [9]
- Steyerberg and Vergouwe (2014) [17]
- Jaeger et al. (2020) [18]

Validate-then-MI

- Musoro et al. (2014) [19]
- Wahl et al. (2016) [20]
- Mertens et al. (2020) [21]
- Carroll (2022) [22]

Recommendations

Validate-then-MI which includes applying CV first and then multiply imputing within each iteration of CV is recommended.

Simulation studies have shown that while *MI-then-CV* has smaller bias for AUC and Brier score when compared to the same simulated data with no missing values. When the prediction model was evaluated in a larger unseen dataset, these methods tended to be over-optimistic.

CV-then-MI had similar performance to *MI-then-CV* when compared to fully-observed data, but was not over-optimistic when evaluated on a larger unseen dataset.

Additional considerations

- For smaller sample sizes with rare or low prevalence outcomes, running CV can occasionally lead to errors. This is because in certain iterations of cross-validation, there are no events in some of the folds and the analyst may need to randomly sample events equally across folds.

Conclusions

Conclusions

- Use sample size calculations to ensure there is an appropriate number of records to train a prediction model
- Use a training and test imputation model. The training imputation model should include all relevant covariates and the outcome. The test imputation model should include all relevant covariates and, potentially, the outcome if the ideal performance is of interest.
- Apply validation algorithm first and multiply impute within (for CV this means *CV-then-MI*)
- When reporting the development of a prediction model, follow guidelines such as TRIPOD reporting guidelines

Reporting

Transparency and reproducibility are important and often hindered by reporting. While it may not be possible to include all relevant information in a paper, the supplementary material should contain key information. Some suggestions are below but reporting guidelines should be additionally consulted:

- Baseline table (may be stratified by training, test and calibration if analyst has selected this method) with each covariate stating the number/percentage of missingness
- Missing data assumptions
- How you have chosen to handle missing data:
- Validate first or impute first
- Choice of M
- What variables have been included in the imputation model

Extra material

Internal validation: Bootstrapping

The bootstrap is another method that is commonly used to internally validate the predictive performance of a clinical prediction model.

A bootstrap sample involves randomly sampling observations (with replacement) from a dataset. The number of observations in each bootstrap sample $b = 1, \ldots, B$ is the same as in the dataset, but the bootstrap sample may have repeated observations.

The basic idea is to sample with replacement from the dataset to train and evaluate a model. In doing so, it is expected that 63.2% of observations are represented at least once in a bootstrap sample [23, p.253].

Internal validation: Bootstrapping

There are several variations of using the bootstrap for correcting the optimism in the apparent performance estimate [24]. The default/standard method and the *0.632* variation.

Evaluating a prediction model P in a dataset $\mathbf{D}$ for a particular performance measure is noted as $\text{perf}(P, \mathbf{D})$.

In the following algorithm a dataset is denoted as $\mathbf{D}_d$ where d represents either the full dataset (o) or the bootstrap sample (b) and a prediction model developed in either the full dataset or bootstrap sample will be labelled P_d for $d = o, b$.

Standard bootstrap with no missing data

The *standard* bootstrap method uses the following steps:

1. Train a prediction model P_o on the full dataset. Evaluate the performance of P_o in the original data to estimate the *apparent performance*, $\text{perf}(P_o, \mathbf{D}_o)$.
2. Take a bootstrap sample b from the original data. Train a prediction model P_b in the bootstrap sample b.
3. Evaluate the performance of P_b in bootstrap sample b to estimate the *bootstrap performance* ($\text{perf}(P_b, \mathbf{D}_b)$).
4. Evaluate the performance of P_b in the original data to estimate the *test performance* ($\text{perf}(P_b, \mathbf{D}_o)$).
5. The *bootstrap performance* (of the prediction model fitted to bootstrap sample b) is then compared to the *test performance*. This produces an estimate of optimism for the *bootstrap performance*. This is estimated as:

$$\text{Optimism}_b = \text{bootstrap performance}_b - \text{test performance}_b$$

6. Steps (2)-(5) are repeated for $b = 1, \ldots, B$.
7. The optimism-corrected performance (OCP) is then calculated as

$$\text{OCP} = \text{Apparent performance} - \frac{1}{B}\sum_{b=1}^{B} \text{Optimism}_b$$

0.632 bootstrap with no missing data

For the *standard* bootstrap algorithm there are three measures of interest: Apparent, Bootstrap (Apparent performance in the Bootstrap sample) and Test.

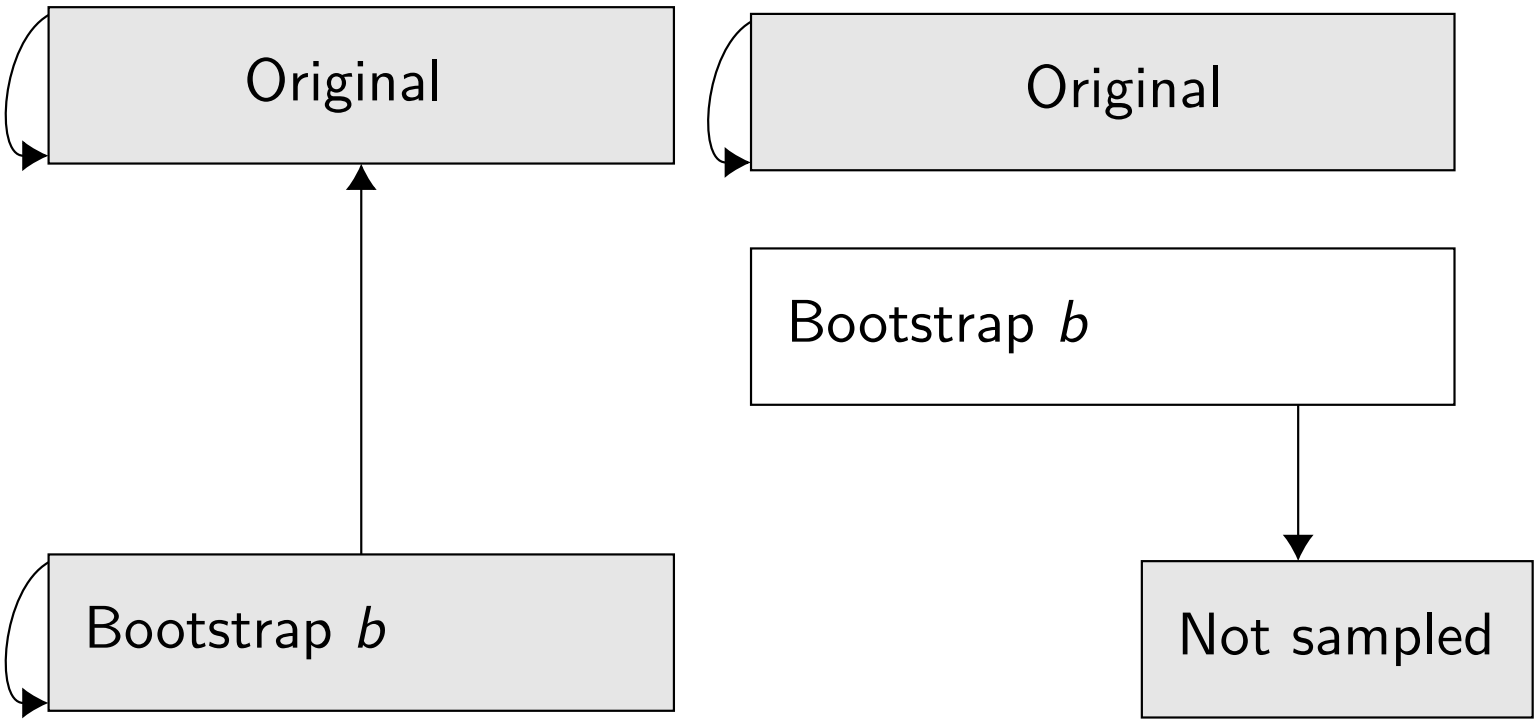
A variation of this validation method is the '0.632' algorithm which is similar to the OOB bootstrap.

The 0.632 algorithm differs from the *standard* bootstrap by evaluating the bootstrap prediction model P_b only in those observations which were not selected in the bootstrap sample in order to estimate the test performance.

This means that Step 3 is ignored and Step 4 is modified to only evaluate P_b in those not bootstrap sampled.

The optimism-corrected performance (OCP) of the 0.632 method is
$(0.368 \times \text{Apparent performance}) + (0.632 \times \text{mean(Test performance)})$
[23, p.253].

Optimism corrected bootstrap with no missing data



(c) *Standard* algorithm **(d)** 0.632 algorithm

The difference between the *standard* and 0.632 algorithms for one bootstrap sample b. The solid lined arrow starts where the prediction model was developed and points to the data where it is evaluated. Datasets coloured in grey are used for the evaluation of a prediction model.

Bootstrapping (BS) with missing data

When conducting the bootstrap, to date, none of the current literature has made clear how they bootstrapped with respect to the missing data.

I recommend using the stratified bootstrap method to ensure the same proportion of observed and missing is present in each bootstrap sample. For smaller sample sizes, the stratified bootstrap avoids the possibility of sampling observations which all contained missing values in the covariates.

Similarly to using CV, we can consider imputing the original data first and then using bootstrap validation (*MI-then-BS*) or we can take a bootstrap sample first and then impute within the bootstrap sample (*BS-then-MI*).

BS and missing data

As in the inference literature [25], *BS-then-MI* is the preferred order for OCP.

Initial work for ideal and pragmatic scenarios found that *BS-then-MI* performed well with both internal validation and with a larger 'unseen' dataset compared to *MI-then-BS* methods [22].

Below, we outline the *standard* and 0.632 methods adapted for *BS-then-MI*.

Standard algorithm with missing data

1. The original sample (o) contains missing data. Impute it using both training and test imputation models to get the *Apparent* performance:
   - **1.1** Use the training imputation model to get M_{train} imputed datasets
   - **1.2** In each of these imputed datasets m_{train}, train the prediction model $P_{m_{train}}$.
   - **1.3** Impute the original sample again but this time using the test imputation model to get M_{test} imputed datasets ($t = 1, \ldots, M_{test}$).
   - **1.4** For each $m_{train} = 1, \ldots, M_{train}$ calculate the prediction model's performance across the test imputed datasets, t, and use Rubin's first rule to get an overall estimate of performance for each model:
   $$\text{perf}(P_{m_{train}}, o) = \sum_{t=1}^{M_{test}} \frac{\text{perf}(P_{m_{train}}, t)}{M_{test}}$$
   - **1.5** Use Rubin's first rule to get an overall estimate of apparent performance
   $$Apparent = \sum_{m=1}^{M_{train}} \frac{\text{perf}(P_{m_{train}}, o)}{M_{train}}$$

2. Sample from the original data with replacement to get a bootstrap sample b. Impute it using both training and test imputation models to get the *Bootstrap* performance:
   - 2.1 Use the training imputation model to get M_{train} imputed datasets of bootstrap sample b.
   - 2.2 For each imputed dataset, $m^*_{train} = 1, \ldots, M_{train}$ of bootstrap sample b, train the prediction model $P_{m^*_{train}}$
   - 2.3 Impute bootstrap sample b using the test imputation model to get M_{test} imputed datasets
   - 2.4 For each $m^*_{train} = 1, \ldots, M_{train}$ calculate the prediction model's performance across the bootstrap test imputed datasets, t^*, and use Rubin's first rule to get an overall estimate of the bootstrap performance for each model:

$$perf(P_{m^*_{train}}, b) = \sum_{t^*=1}^{M_{test}} \frac{\text{perf}(P_{m^*_{train}}, t^*)}{M_{test}}$$

   - 2.5 Use Rubin's first rule to get an overall estimate of the bootstrap performance for bootstrap sample b

$$Bootstrap_b = \sum_{m^*_{train}=1}^{M_{train}} \frac{\text{perf}(P_{m^*_{train}}, b)}{M_{train}}$$

3. Finally, to get the *Test* performance:
   3.1 Impute the original sample using the test imputation model to get a different set of M_{test} imputed datasets ($t' = 1, \ldots, M_{test}$) to those in step (3)
   3.2 For each bootstrap prediction model from step (2) $P_{m^*_{train}}$ for $m^*_{train} = 1, \ldots, M_{train}$, calculate the performance across the test imputed datasets $t' = 1, \ldots, M_{test}$ from step (1) and use Rubin's first rule to get an overall estimate of the test performance for each bootstrap model:

$$perf(P_{m^*_{train}}, o) = \sum_{t'=1}^{M_{test}} \frac{\text{perf}(P_{m^*_{train}}, t')}{M_{test}}$$

   3.3 Use Rubin's first rule to get an overall estimate of the test performance across the M_{train} prediction models

$$Test_b = \sum_{m^*_{train}=1}^{M_{train}} \frac{\text{perf}(P_{m^*_{train}}, o)}{M_{train}}$$

5. The optimism is then estimated as the difference between the bootstrap and test performance:

$$Optimism_b = Bootstrap_b - Test_b$$

6. Repeat steps (2) - (5) B times
7. To estimate the optimism-corrected performance,

$$OCP = Apparent - \frac{1}{B}\sum_{b=1}^{B} Optimism_b$$

0.632 variation with missing data

In this variation of the bootstrap algorithm only the apparent and test performance are of interest. The 0.632 algorithm when bootstrapping first, followed by MI, has similar steps to the *standard* algorithm above but with some minor changes as noted below:

- The bootstrap performance is not calculated (because the 0.632 version is using a more classic sample splitting approach) so steps (3) - (5) are not necessary
- For step (1) take those observations not sampled in the bootstrap sample from the original sample. Impute this subsample of not selected observations using the test imputation model to get $t' = 1, \ldots, M_{test}$ imputed datasets.
- For step (7) the optimism-corrected performance becomes a weighted average (recall that on average, approximately 63.2% of observations in a bootstrap sample are unique) of the apparent performance and B test performances

$$OCP = (0.368 \times Apparent) + \left(0.632 \times \frac{1}{B} \sum_{b=1}^{B} Test_b\right)$$

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